

# AutoNetKit User Manual

## Type-Safe Network Topology Modelling

---

Simon Knight

Adelaide, Australia

March 2026

**About this manual.** This manual covers installing AutoNetKit, building typed network topologies with Pydantic models, deriving multi-layer protocol views, querying topology data with the fluent API, running design-rule validation before configuration generation, and compiling validated designs into vendor-specific device configurations. Chapters on the layer system, the query DSL, built-in design rules, supported output platforms, and ready-made batteries\_included topologies make this the primary practitioner reference for AutoNetKit 2.1.0.

## Contents

|          |                                            |           |
|----------|--------------------------------------------|-----------|
| <b>1</b> | <b>Quick Start</b>                         | <b>1</b>  |
| <b>2</b> | <b>Installation and Prerequisites</b>      | <b>2</b>  |
| 2.1      | Requirements . . . . .                     | 2         |
| 2.2      | Installing from PyPI . . . . .             | 2         |
| 2.3      | Verifying the Installation . . . . .       | 2         |
| 2.4      | Upgrading . . . . .                        | 3         |
| <b>3</b> | <b>Core Concepts</b>                       | <b>3</b>  |
| <b>4</b> | <b>Core Workflows</b>                      | <b>3</b>  |
| 4.1      | Defining Custom Models . . . . .           | 3         |
| 4.2      | Building the Physical Topology . . . . .   | 4         |
| 4.3      | Deriving Protocol Layers . . . . .         | 5         |
| 4.4      | Querying Topology Data . . . . .           | 6         |
| 4.5      | Running Design Rules . . . . .             | 7         |
| 4.6      | Generating Device Configurations . . . . . | 7         |
| <b>5</b> | <b>Layer Configuration Reference</b>       | <b>8</b>  |
| 5.1      | Parameters . . . . .                       | 8         |
| 5.2      | Common Patterns . . . . .                  | 8         |
| 5.2.1    | Full Mesh (iBGP) . . . . .                 | 8         |
| 5.2.2    | Same-AS OSPF . . . . .                     | 8         |
| 5.2.3    | Cross-AS eBGP . . . . .                    | 8         |
| 5.2.4    | Ring Topology . . . . .                    | 8         |
| 5.2.5    | Hub and Spoke . . . . .                    | 9         |
| 5.3      | Cross-Layer Traversal . . . . .            | 9         |
| <b>6</b> | <b>Python API Quick Reference</b>          | <b>9</b>  |
| 6.1      | Node and Endpoint Management . . . . .     | 9         |
| 6.2      | Edge and Link Management . . . . .         | 9         |
| 6.3      | Layer Management . . . . .                 | 10        |
| 6.4      | Query API . . . . .                        | 10        |
| 6.5      | IO and Serialisation . . . . .             | 11        |
| <b>7</b> | <b>Design Rules Catalogue</b>              | <b>11</b> |
| 7.1      | Built-In Rules . . . . .                   | 11        |
| 7.2      | Rule Composition Operators . . . . .       | 12        |
| 7.3      | Writing a Custom Rule . . . . .            | 13        |
| <b>8</b> | <b>Configuration Generation</b>            | <b>13</b> |
| 8.1      | Supported Platforms . . . . .              | 14        |
| 8.2      | Compiler Usage . . . . .                   | 14        |
| 8.3      | Jinja2 Template Customisation . . . . .    | 14        |
| 8.4      | Deployment Environment Export . . . . .    | 14        |
| <b>9</b> | <b>batteries_included Topologies</b>       | <b>15</b> |
| 9.1      | Available Factories . . . . .              | 15        |
| 9.1.1    | Leaf-Spine Datacenter . . . . .            | 15        |
| 9.1.2    | ISP Internet Exchange . . . . .            | 15        |
| 9.1.3    | Enterprise Campus . . . . .                | 15        |

---

|           |                                                            |           |
|-----------|------------------------------------------------------------|-----------|
| 9.1.4     | WAN Mesh . . . . .                                         | 15        |
| 9.2       | Extending a Factory Topology . . . . .                     | 16        |
| <b>10</b> | <b>Troubleshooting</b>                                     | <b>16</b> |
| <b>11</b> | <b>Integration with the Ecosystem</b>                      | <b>17</b> |
| 11.1      | NTE – Graph Storage and Queries . . . . .                  | 17        |
| 11.2      | NetCfg – Rust-Native Configuration Compiler . . . . .      | 17        |
| 11.3      | TopoGen – Seed-Deterministic Topology Generation . . . . . | 18        |
| 11.4      | netsim / netlab – Network Simulation . . . . .             | 18        |
| 11.5      | NetVis – Topology Visualisation . . . . .                  | 18        |
| 11.6      | Workbench – Orchestration . . . . .                        | 18        |
| <b>12</b> | <b>Further Reading</b>                                     | <b>19</b> |

# 1 Quick Start

Go from zero to a running, validated topology in under ten minutes.

**Step 1.** Install AutoNetKit (the NTE Rust engine installs automatically):

```
pip install autonetkit
```

**Step 2.** Create and populate a topology:

```
from pydantic import BaseModel
from autonetkit import Topology
from autonetkit.models.base import BaseTopologyNode, BaseTopologyEndpoint
from autonetkit.models.edge import BaseInternodeEdge

class RouterData(BaseModel):
    label: str
    asn: int = 65000
    platform: str = "cisco_ios"

class Router(BaseTopologyNode[RouterData]):
    pass

class InterfaceData(BaseModel):
    label: str

class Interface(BaseTopologyEndpoint[InterfaceData]):
    pass

class LinkData(BaseModel):
    bandwidth: int | None = None

class Link(BaseInternodeEdge[LinkData]):
    type: str = "link"
```

**Step 3.** Build the physical topology:

```
topo = Topology()
r1 = Router(layer="physical", data=RouterData(label="R1"))
r2 = Router(layer="physical", data=RouterData(label="R2"))
topo.nodes.add([r1, r2])

# Add interfaces and connect them
i1 = topo.endpoint_mgr.add(
    Interface, layer="physical",
    data=InterfaceData(label="Eth0/0"), parent_id=r1.id,
)
i2 = topo.endpoint_mgr.add(
    Interface, layer="physical",
    data=InterfaceData(label="Eth0/0"), parent_id=r2.id,
)
topo.edges.add(Link(data=LinkData(bandwidth=10000), src=i1, dst=i2))
```

**Step 4.** Derive protocol layers, validate, and compile:

```
# Derive an OSPF layer
```

```

topo.layers.build({"ospf": {"parent": "physical",
                           "keep_edges_where_same": "asn"}})

# Validate the design
from autonetkit.analysis import standard_ruleset
report = standard_ruleset().run(topo)
print(report.to_text())

# Compile to IOS configuration
from autonetkit.compilers.registry import get_compiler
compiler = get_compiler("ios")
config = compiler.compile(topo, r1.id)
print(compiler.render_config(config))

```

**Step 5.** Query node counts per layer:

```

# In a Python REPL or script
for name in topo.layers.names():
    print(name, topo.query.nodes().in_layer(name).count())

```

## 2 Installation and Prerequisites

### 2.1 Requirements

#### ▷ Prerequisites

- **Python 3.9 or later** (3.11+ recommended for best performance).
- **Pydantic v2** — installed automatically as a dependency.
- **NTE engine** (`ank_nte`) — the Rust graph-storage backend; installed automatically as a dependency.
- **Polars** — columnar query execution; installed automatically.
- An active Python virtual environment is strongly recommended.

### 2.2 Installing from PyPI

```
pip install autonetkit
```

#### Tip

Use `uv pip install autonetkit` for significantly faster dependency resolution in projects with large dependency trees. `uv` resolves the Rust wheel for your platform in a single round-trip rather than iterating through `pip`'s backtracking solver.

#### Caution

The NTE engine (`ank_nte`) ships pre-built wheels for Linux (x86\_64, aarch64), macOS (Intel, Apple Silicon), and Windows (x86\_64). If you are on an unsupported platform, `pip` will attempt a source build. This requires a stable Rust toolchain (`rustup`) and the `maturin` build tool. Install Rust first with:

```
curl -proto '=https' -tlsv1.2 -sSf https://sh.rustup.rs | sh
```

### 2.3 Verifying the Installation

```
python -c "import autonetkit; print(autonetkit.__version__)"
python -c "import ank_nte; print(ank_nte.__version__)"
```

Both commands should print `2.1.0` and the NTE version string respectively.

### 2.4 Upgrading

```
pip install --upgrade autonetkit
```

Caution

AutoNetKit 2.x uses a different NDJSON serialisation protocol to 1.x. Topology objects serialised under 1.x cannot be loaded by 2.x directly. Use `autonetkit.io.migrate_v1(path)` to convert legacy NDJSON exports before loading them into a 2.x Topology.

## 3 Core Concepts

AutoNetKit organises a network topology into three kinds of element:

- Nodes** represent devices or logical network elements: routers, switches, hosts, VLANs, VRFs. Nodes are first-class graph citizens.
- Endpoints** represent connection points on devices: physical interfaces, loopbacks, sub-interfaces. Each endpoint belongs to exactly one node via an ownership edge.
- Edges** represent relationships between elements. Three edge kinds enforce strict connection rules: *internode* (endpoint to endpoint across devices), *intranode* (node to node within a device), and *ownership* (any node to any node, containment).

Every element carries a generic data: `T` field backed by a user-defined Pydantic `BaseModel`. This gives full IDE autocompletion, runtime validation, and free serialisation — with no stringly-typed dictionaries.

The Topology façade exposes managers for every domain:

### Command Reference

| Command                            | Description                                      |
|------------------------------------|--------------------------------------------------|
| <code>topology.nodes</code>        | Add, remove, and register node models            |
| <code>topology.endpoint_mgr</code> | Add and query endpoints (interfaces)             |
| <code>topology.edges</code>        | Add, remove, and query edges                     |
| <code>topology.links</code>        | Simplified link-level (node-to-node) view        |
| <code>topology.layers</code>       | Create, derive, freeze, and query layers         |
| <code>topology.query</code>        | Fluent query API over nodes, edges, links, paths |
| <code>topology.io</code>           | Serialise / deserialise (NDJSON, JSON, GraphML)  |
| <code>topology.analysis</code>     | Run rule sets and retrieve reports               |

The Topology class is a thin façade wiring approximately eighteen domain managers over a shared NTE store. Use dot-completion to explore the API.

## 4 Core Workflows

### 4.1 Defining Custom Models

Every domain-specific element type inherits from one of AutoNetKit’s base classes parameterised by a data model:

**Step 1.** Define a Pydantic data class for each element kind:

```
from pydantic import BaseModel

class RouterData(BaseModel):
    label: str
```

Inherit `BaseTopologyNode` for devices and `BaseTopologyEndpoint` for interfaces. Both are generic over a Pydantic `BaseModel` subclass.

```

    role: str = "core"          # "core" | "edge" | "spine" | "leaf"
    asn: int = 65000
    platform: str = "cisco_ios"
    loopback: str | None = None

class InterfaceData(BaseModel):
    label: str
    speed: str | None = None    # e.g. "10G", "100G"
    ip: str | None = None

class LinkData(BaseModel):
    bandwidth: int | None = None # Mbps
    metric: int = 1

```

**Step 2.** Inherit the appropriate base class, parameterised by your data model:

```

from autonetkit.models.base import (
    BaseTopologyNode, BaseTopologyEndpoint,
)
from autonetkit.models.edge import BaseInternodeEdge

class Router(BaseTopologyNode[RouterData]):
    pass # inherits id, layer, data; extend for domain logic

class Interface(BaseTopologyEndpoint[InterfaceData]):
    pass

class Link(BaseInternodeEdge[LinkData]):
    type: str = "link" # discriminator used by registry

```

**Step 3.** Register models with the topology so the identity map can hydrate them:

```

topo = Topology()
topo.nodes.register_models([Router, Interface])
topo.edges.register_models([Link])

```

### Tip

Use `FlexibleData` (from `autonetkit.models.base`) as the data type during rapid prototyping. It sets `extra="allow"` so any field is accepted without a formal schema. Migrate to a typed model before running design rules or generating configurations.

## 4.2 Building the Physical Topology

**Step 1.** Instantiate a Topology and add nodes:

```

from autonetkit import Topology

topo = Topology()

core1 = Router(layer="physical",
               data=RouterData(label="Core1", role="core"))
core2 = Router(layer="physical",
               data=RouterData(label="Core2", role="core"))
topo.nodes.add([core1, core2])
# IDs are assigned by NTE after add(); router.id is now set

```

**Step 2.** Add endpoints (interfaces) owned by each node:

```
# endpoint_mgr.add() creates the endpoint and the ownership edge
c1_eth0 = topo.endpoint_mgr.add(
    Interface, layer="physical",
    data=InterfaceData(label="Eth0/0", speed="10G"),
    parent_id=core1.id,
)
c2_eth0 = topo.endpoint_mgr.add(
    Interface, layer="physical",
    data=InterfaceData(label="Eth0/0", speed="10G"),
    parent_id=core2.id,
)
```

**Step 3.** Connect endpoints with an internode edge (physical link):

```
link = Link(
    data=LinkData(bandwidth=10000, metric=1),
    src=c1_eth0,
    dst=c2_eth0,
)
topo.edges.add(link)
```

**Step 4.** Verify the topology:

```
print(topo.query.nodes().in_layer("physical").count()) # 2
print(topo.query.edges().in_layer("physical").count()) # 1
```

**Caution**

Node IDs are assigned by NTE *after* calling `topo.nodes.add()`. Do not capture `node.id` before the node has been added — it will be `None`. Batch-add is more efficient than adding nodes one by one for topologies with more than a few hundred elements.

## 4.3 Deriving Protocol Layers

Protocol views are derived from the physical layer using a configuration dictionary:

**Step 1.** Define the layer configuration:

```
LAYERS = {
    "ospf": {
        "parent": "physical",
        # Clone edges only where both endpoints share the same ASN
        "keep_edges_where_same": "asn",
    },
    "ebgp": {
        "parent": "physical",
        # Clone edges only where endpoints cross AS boundaries
        "keep_edges_where_different": "asn",
    },
    "ibgp": {
        "parent": "physical",
        "copy_edges": False,
        # Group nodes by ASN; create a full mesh within each group
        "group_by": "asn",
        "create": "full_mesh",
    },
}
```

*Layer derivation is declarative, not reactive. Call `layers.build()` after all physical nodes and links are in place.*



**Step 2. Validate and build:**

```
errors = topo.layers.validate(LAYERS)
if errors:
    for e in errors:
        print("Layer config error:", e)
else:
    topo.layers.build(LAYERS)
```

**Step 3. Optionally freeze source layers to prevent accidental mutation:**

```
topo.layers.freeze("physical")
assert topo.layers.frozen("physical")
```

## 4.4 Querying Topology Data

AutoNetKit's query API is fluent and type-safe:

**Step 1. Query all nodes of a type in a layer:**

```
# List[Router] -- fully hydrated Pydantic models
core_routers = (
    topo.query.nodes()
    .of_type(Router)
    .in_layer("physical")
    .where(role="core")
    .models()
)
```

*Queries are lazy builders. Terminal methods (.models(), .ids(), .count()) execute against Rust.*

**Step 2. Use expression filters for complex predicates:**

```
from autonetkit import q

# Field comparison
high_bw_links = (
    topo.query.links()
    .in_layer("physical")
    .filter(q.field("bandwidth") >= 100000)
    .collect()
)

# String match + logical AND
cisco_core = (
    topo.query.nodes()
    .of_type(Router)
    .filter(
        (q.field("platform").contains("cisco")) &
        (q.field("role") == "core")
    )
    .models()
)
```

**Step 3. Use terminal methods suited to the task:**

```
count = topo.query.nodes().of_type(Router).count()
exists = topo.query.nodes().where(role="spine").exists()
```

```
first = topo.query.nodes().of_type(Router).first()
one   = topo.query.nodes().where(label="Core1").one()
df    = topo.query.nodes().of_type(Router).dataframe()
```

**Step 4.** Navigate across layers:

```
# Find the plan-layer counterpart of a physical router
plan_router = physical_router.in_plan(topo)

# Find the same router in the OSPF layer
ospf_router = physical_router.in_layer("ospf", topo)
```

## 4.5 Running Design Rules

**Step 1.** Run the built-in standard ruleset:

```
from autonetkit.analysis import standard_ruleset

report = standard_ruleset().run(topo)
print(report.to_text(verbose=True))
```

**Step 2.** Check the result and act on failures:

```
if not report.passed:
    print(f"Errors: {report.error_count}, "
          f"Warnings: {report.warning_count}")
    for r in report.filter_failed():
        print(f" [{r.severity.value.upper()}] {r.message}")
    raise SystemExit(1)
```

**Step 3.** Export the report as a Polars DataFrame for CI integration:

```
df = report.to_dataframe()
df.write_csv("validation_report.csv")
```

## 4.6 Generating Device Configurations

**Step 1.** Look up the compiler for a platform:

```
from autonetkit.compilers.registry import get_compiler

compiler = get_compiler("ios")          # Cisco IOS
# or: get_compiler("eos"), get_compiler("junos"), ...
```

**Step 2.** Compile and render for each device:

```
for router in topo.query.nodes().of_type(Router).models():
    compiler = get_compiler(router.data.platform)
    config = compiler.compile(topo, router.id)
    cli_text = compiler.render_config(config)
    out_path = f"output/{router.data.label}.cfg"
    with open(out_path, "w") as f:
        f.write(cli_text)
```

**Step 3.** Export for a deployment environment:

```

from autonetkit.export import ContainerlabExporter

exporter = ContainerlabExporter()
exporter.export(topo, output_dir="lab/")
# Produces lab/topology.clab.yml and per-device config files

```

## 5 Layer Configuration Reference

A layer configuration dictionary maps layer names to parameter objects. The build system resolves dependencies, validates the configuration, and builds layers in topological order.

### 5.1 Parameters

#### Layer Parameters

| Command                    | Description                                                    |
|----------------------------|----------------------------------------------------------------|
| parent                     | Name of the parent layer to clone nodes from                   |
| keep_edges_where_same      | Clone edges only where data.<field> matches on both ends       |
| keep_edges_where_different | Clone edges only where data.<field> differs                    |
| copy_edges                 | Boolean; if False, no edges are copied (default: True)         |
| group_by                   | Field name; groups nodes by field value for topology creation  |
| create                     | Topology pattern within each group: full_mesh, hub_spoke, ring |

### 5.2 Common Patterns

#### 5.2.1 Full Mesh (iBGP)

Nodes grouped by ASN; a full mesh of virtual links is created within each group. This models iBGP sessions without requiring explicit physical links.

```

ibgp:
  parent: physical
  copy_edges: false
  group_by: asn
  create: full_mesh

```

#### 5.2.2 Same-AS OSPF

Only physical links between routers in the same AS are retained.

```

ospf:
  parent: physical
  keep_edges_where_same: asn

```

#### 5.2.3 Cross-AS eBGP

Only physical links crossing AS boundaries are retained.

```

ebgp:
  parent: physical
  keep_edges_where_different: asn

```

#### 5.2.4 Ring Topology

Group nodes by a field and connect them in a ring (useful for modelling ring access networks).

```
access_ring:
  parent: physical
  copy_edges: false
  group_by: region
  create: ring
```

### 5.2.5 Hub and Spoke

Group nodes by a field; the node with the highest degree within each group becomes the hub.

```
hub_spoke:
  parent: physical
  copy_edges: false
  group_by: pop
  create: hub_spoke
```

## 5.3 Cross-Layer Traversal

After derivation, every cloned node carries pointers back to its parent:

```
# Get the physical counterpart of an OSPF node
ospf_router = topo.query.nodes().of_type(Router) \
    .in_layer("ospf").first()

physical_router = ospf_router.in_plan(topo)      # -> plan layer
ibgp_router     = ospf_router.in_layer("ibgp", topo) # -> iBGP layer
```

## 6 Python API Quick Reference

AutoNetKit is primarily a library. The following tables summarise the most commonly used manager methods.

### 6.1 Node and Endpoint Management

#### NodeManager — `topology.nodes`

| Command                              | Description                                  |
|--------------------------------------|----------------------------------------------|
| <code>.add(nodes)</code>             | Add a list of nodes; assigns IDs in place    |
| <code>.remove(node_id)</code>        | Remove node and all its edges                |
| <code>.get(node_id)</code>           | Retrieve a node by ID                        |
| <code>.register_models(types)</code> | Register model classes with the identity map |
| <code>.count()</code>                | Total node count across all layers           |

#### EndpointManager — `topology.endpoint_mgr`

| Command                                        | Description                        |
|------------------------------------------------|------------------------------------|
| <code>.add(cls, layer, data, parent_id)</code> | Create endpoint and ownership edge |
| <code>.get(ep_id)</code>                       | Retrieve endpoint by ID            |
| <code>.of_node(node_id)</code>                 | List all endpoints owned by a node |

### 6.2 Edge and Link Management

#### EdgeManager — `topology.edges`

| Command                       | Description                           |
|-------------------------------|---------------------------------------|
| <code>.add(edge)</code>       | Add a typed edge; assigns ID in place |
| <code>.remove(edge_id)</code> | Remove edge                           |
| <code>.get(edge_id)</code>    | Retrieve edge by ID                   |
| <code>.count()</code>         | Total edge count                      |

### LinkManager — `topology.links`

| Command                           | Description                  |
|-----------------------------------|------------------------------|
| <code>.between(a_id, b_id)</code> | Find links between two nodes |
| <code>.neighbours(node_id)</code> | List neighbour node IDs      |
| <code>.degree(node_id)</code>     | Connection degree of a node  |

## 6.3 Layer Management

### LayerManager — `topology.layers`

| Command                           | Description                              |
|-----------------------------------|------------------------------------------|
| <code>.build(config)</code>       | Build layers from config dict            |
| <code>.validate(config)</code>    | Validate config dict; returns error list |
| <code>.get(name)</code>           | Return a LayerHandle                     |
| <code>.names()</code>             | List all layer names                     |
| <code>.freeze(name)</code>        | Prevent further mutation                 |
| <code>.frozen(name)</code>        | Check frozen status                      |
| <code>.clone_to_layer(...)</code> | Clone a node set to a new layer          |

## 6.4 Query API

### QueryManager — `topology.query`

| Command               | Description        |
|-----------------------|--------------------|
| <code>.nodes()</code> | Start a NodeQuery  |
| <code>.edges()</code> | Start an EdgeQuery |
| <code>.links()</code> | Start a LinkQuery  |
| <code>.paths()</code> | Start a PathQuery  |

### NodeQuery chaining

| Command                       | Description                                    |
|-------------------------------|------------------------------------------------|
| <code>.of_type(Model)</code>  | Type-narrowing filter                          |
| <code>.in_layer(name)</code>  | Layer filter                                   |
| <code>.where(**fields)</code> | Equality filter on <code>data.*</code> fields  |
| <code>.filter(*exprs)</code>  | Expression-based filter using <code>q.*</code> |
| <code>.to_plan()</code>       | Navigate to plan-layer counterparts            |
| <code>.to_layer(name)</code>  | Navigate to another layer                      |

### NodeQuery terminals

| Command                   | Description                                                 |
|---------------------------|-------------------------------------------------------------|
| <code>.models()</code>    | <code>List[T]</code> of hydrated Pydantic models            |
| <code>.ids()</code>       | <code>List[int]</code> of node IDs                          |
| <code>.count()</code>     | <code>int</code> (no model materialisation)                 |
| <code>.first()</code>     | <code>Optional[T]</code> — first match or <code>None</code> |
| <code>.one()</code>       | <code>T</code> — exactly one match (raises if 0 or 2+)      |
| <code>.exists()</code>    | <code>bool</code>                                           |
| <code>.dataframe()</code> | <code>pl.DataFrame</code> (Polars)                          |

## 6.5 IO and Serialisation

### IOManager — `topology.io`

| Command                        | Description                           |
|--------------------------------|---------------------------------------|
| <code>.save(path)</code>       | Serialise topology to NDJSON          |
| <code>.load(path)</code>       | Deserialise from NDJSON               |
| <code>.to_json()</code>        | Return JSON string                    |
| <code>.to_graphml(path)</code> | Export to GraphML (for visualisation) |
| <code>.migrate_v1(path)</code> | Convert 1.x NDJSON to 2.x format      |

## 7 Design Rules Catalogue

The rule engine is AutoNetKit’s “network linter”: composable, declarative validation that runs before configuration generation, catching design errors before they become operational incidents.

### 7.1 Built-In Rules

AutoNetKit ships 27 built-in rules across five categories. The `standard_ruleset()` factory runs the four most critical.

| Category    | Rule ID                      | Purpose                               | Severity |
|-------------|------------------------------|---------------------------------------|----------|
| Validation  | validation.isolated_nodes    | Nodes with no connections             | ERROR    |
|             | validation.required_field    | Mandatory field is set                | ERROR    |
|             | validation.min_connections   | Minimum degree per node type          | ERROR    |
|             | validation.unique_labels     | No duplicate node labels              | WARNING  |
|             | validation.no_self_loops     | Edges do not connect a node to itself | ERROR    |
| Anomaly     | anomaly.spoof                | Articulation points (single PoF)      | WARNING  |
|             | anomaly.disconnected         | Partitioned topology components       | ERROR    |
|             | anomaly.high_degree          | Nodes above degree threshold          | WARNING  |
|             | anomaly.orphaned_endpoint    | Endpoints not connected to any link   | WARNING  |
|             | anomaly.stub_asn             | ASN with only one external link       | INFO     |
|             | anomaly.duplicate_link       | Parallel links between same pair      | INFO     |
| Pattern     | pattern.hub_spoke            | Star topology detected                | INFO     |
|             | pattern.full_mesh            | Fully connected clique                | INFO     |
|             | pattern.ring                 | Ring topology detected                | INFO     |
|             | pattern.asymmetric_degree    | Unbalanced degree distribution        | WARNING  |
|             | pattern.triangle_free        | No triangles in physical layer        | INFO     |
| Resilience  | resilience.dual_homed        | Nodes with $\geq 2$ upstreams         | WARNING  |
|             | resilience.path_diversity    | Edge-disjoint path count              | WARNING  |
|             | resilience.link_redundancy   | Redundant links per pair              | INFO     |
|             | resilience.min_upstreams     | Minimum upstream count                | ERROR    |
|             | resilience.max_radius        | Diameter bound                        | WARNING  |
|             | resilience.ecmp_capacity     | Equal-cost path count                 | INFO     |
| Operational | operational.missing_attr     | Required attribute absent             | ERROR    |
|             | operational.asymmetric_link  | Speed mismatch across a link          | WARNING  |
|             | operational.platform_support | Platform has a registered compiler    | ERROR    |
|             | operational.loopback_missing | Routers without a loopback address    | WARNING  |
|             | operational.asn_conflict     | Same ASN used by unrelated routers    | ERROR    |

## 7.2 Rule Composition Operators

Five operators build complex rules from simple leaf rules:

```

from autonetkit.analysis import AllOf, AnyOf, IfThen, Not, AtLeast

# All listed rules must pass
mandatory = AllOf(
    IsolatedNodesRule(),
    MinConnectionsRule(Router, min_count=2),
    rule_id="mandatory_checks",
)

# Conditional: only check dual-homing on core routers
core_check = IfThen(
    condition=NodeTypeRule(CoreRouter),
    consequence=DualHomedRule(CoreRouter, min_upstreams=3),
    rule_id="core_redundancy",
)

# At least 2 of 3 resilience rules must pass
resilience = AtLeast(2,
    PathDiversityRule(min_paths=2),
    LinkRedundancyRule(),

```

```
DualHomedRule(Router, min_upstreams=2),
rule_id="resilience_floor",
)
```

### 7.3 Writing a Custom Rule

```
from autonetkit.analysis.base import Rule, RuleResult, Severity

class MinInterfacesRule(Rule):
    id = "custom.min_interfaces"
    name = "Minimum Interfaces"
    description = "Each router must have at least N interfaces."
    tags = {"validation", "connectivity"}
    default_severity = Severity.ERROR

    def __init__(self, min_count: int = 2):
        self.min_count = min_count

    def check(self, ctx) -> list[RuleResult]:
        results = []
        for router in ctx.nodes_of_type(Router):
            degree = ctx.get_degree(router.id)
            results.append(RuleResult(
                rule_id = self.id,
                passed = degree >= self.min_count,
                severity = self.default_severity,
                message = (f"{router.data.label} has {degree} "
                           f"interfaces (minimum: {self.min_count})"),
                affected_nodes = [router.id] if degree < self.min_count else [],
            ))
        return results

# Register and run
from autonetkit.analysis import RuleSet
ruleset = RuleSet("custom_checks")
ruleset.add(MinInterfacesRule(min_count=2))
report = ruleset.run(topo)
```

#### Tip

Tag rules with category strings ("validation", "resilience", etc.) and use `report.results_by_tag()` to group the report output by category in CI dashboards.

## 8 Configuration Generation



## 8.1 Supported Platforms

| Vendor    | Platform ID | Compiler Class  | Protocols                  |
|-----------|-------------|-----------------|----------------------------|
| Cisco     | ios         | IOSCompiler     | OSPF, BGP, ISIS, MPLS      |
| Cisco     | iosxr       | IOSXRCompiler   | OSPF, BGP, ISIS, SR, BFD   |
| Cisco     | nxos        | NXOSCompiler    | OSPF, BGP, EVPN, VXLAN     |
| Juniper   | junos       | JunOSCompiler   | OSPF, BGP, ISIS, MPLS, SR  |
| Arista    | eos         | EOSCompiler     | OSPF, BGP, EVPN, VXLAN     |
| Nokia     | sros        | SROSCompiler    | OSPF, BGP, ISIS, SR, L3VPN |
| Nokia     | srlinux     | SRLinuxCompiler | OSPF, BGP, EVPN            |
| FRRouting | frr         | FRRCompiler     | OSPF, BGP, ISIS            |
| SONiC     | sonic       | SONiCCompiler   | BGP, EVPN                  |
| VyOS      | vyos        | VyOSCompiler    | OSPF, BGP                  |
| Cumulus   | cumulus     | CumulusCompiler | OSPF, BGP, EVPN            |

## 8.2 Compiler Usage

```
from autonetkit.compilers.registry import get_compiler, list_platforms

# Discover available platforms
print(list_platforms())

# Compile a single device
compiler = get_compiler("iosxr")
config = compiler.compile(topo, device_id=core1.id)
cli_text = compiler.render_config(config)

# Compile all devices, routing by platform field
for router in topo.query.nodes().of_type(Router).models():
    c = get_compiler(router.data.platform)
    txt = c.render_config(c.compile(topo, router.id))
    print(f"=== {router.data.label} ===\n{txt}")
```

## 8.3 Jinja2 Template Customisation

Each compiler resolves templates from a search path. Drop overrides into a local templates/<platform>/ directory and set the search path:

```
compiler = get_compiler("ios")
compiler.add_template_dir("templates/ios")
# Local templates take precedence over built-in ones
```

Templates receive the compiled config object as context. Variables available in all templates:

```
# Available in every Jinja2 template:
hostname:      "Core1"
platform:      "cisco_ios"
interfaces:    [ {label: "Eth0/0", ip: "10.0.0.1/30", speed: "10G"} ]
ospf_areas:    [ {area: 0, interfaces: ["Eth0/0"]} ]
bgp_neighbors: [ {peer_ip: "10.0.0.2", remote_asn: 65002} ]
```

### Caution

Not every compiler covers every protocol. If a router's topology data includes BGP configuration but the target compiler does not have a BGP template, the rendered configuration will silently omit BGP. Run the `operational.platform_support` design rule before compiling to catch coverage gaps early.

## 8.4 Deployment Environment Export

```
# Containerlab (container-based emulation)
from autonetkit.export import ContainerlabExporter
ContainerlabExporter().export(topo, output_dir="lab/")
# Produces: lab/topology.clab.yml

# Ansible (configuration management)
from autonetkit.export import AnsibleExporter
AnsibleExporter().export(topo, output_dir="ansible/")
# Produces: ansible/inventory.yml + playbooks/

# netsim / netlab (network simulation)
from autonetkit.export import NetsimExporter
NetsimExporter().export(topo, output_dir="sim/")
# Produces: sim/topology.yml
```

## 9 batteries\_included Topologies

The `autonetkit.batteries_included` module ships four ready-made topology factories for common use cases. Each factory returns a fully populated `Topology` with typed models, physical links, and derived protocol layers.

### 9.1 Available Factories

```
from autonetkit.batteries_included import (
    datacenter_topology,
    isp_topology,
    campus_topology,
    wan_topology,
)
```

#### 9.1.1 Leaf-Spine Datacenter

```
topo = datacenter_topology(
    spines=2,      # spine router count
    leaves=4,      # leaf switch count
    asn=65100,
    platform="arista_eos",
)
# Layers: physical, evpn (full mesh between leaves via spines)
```

#### 9.1.2 ISP Internet Exchange

```
topo = isp_topology(
    cores=2,
    edges=4,
    route_servers=1,
    asn_base=65000, # ASN assigned sequentially
    platform="cisco_iosxr",
)
# Layers: physical, ospf (same-ASN), ebgp (cross-ASN), ibgp (full mesh)
```

#### 9.1.3 Enterprise Campus

```
topo = campus_topology(
    core_switches=2,
    distribution_switches=4,
    access_switches=8,
    platform="cisco_ios",
)
# Layers: physical, stp, ospf
```

#### 9.1.4 WAN Mesh

```
topo = wan_topology(
```

```

nodes=8,
seed=42,                # deterministic layout
topology="partial_mesh", # or "ring", "hub_spoke"
platform="junos",
)
# Layers: physical, ospf, ibgp (full mesh)

```

### Tip

All battery-included factories accept a seed parameter for deterministic topology generation. Use the same seed in CI to ensure reproducible test scenarios.

## 9.2 Extending a Factory Topology

Factory topologies are ordinary Topology objects; you can add nodes, edges, or new layers after construction:

```

topo = isp_topology(cores=2, edges=4, asn_base=65000)

# Add a route reflector
rr = Router(layer="physical",
            data=RouterData(label="RR1", role="rr", asn=65000))
topo.nodes.add([rr])

# Re-derive the iBGP layer to include the new node
topo.layers.build({"ibgp": {"parent": "physical",
                           "copy_edges": False,
                           "group_by": "asn",
                           "create": "full_mesh"}})

```

## 10 Troubleshooting

**Problem 1:** NTE compilation error: NteSerialiseError: invalid NDJSON

**Cause:** A Pydantic model contains a field type that the NDJSON codec does not support (e.g., a nested model with circular references, or a custom type without a registered serialiser).

**Solution:** Add a model\_serializer to the offending Pydantic class, or convert the field to a JSON-safe type (str, int, dict). Run model.model\_dump\_json() outside of AutoNetKit first to verify the model serialises correctly.

**Problem 2:** Slow performance when adding thousands of nodes one by one

**Cause:** Each individual topo.nodes.add([node]) call crosses the Python–Rust boundary via NDJSON serialisation. For large topologies this overhead dominates.

**Solution:** Batch all nodes into a single list and call topo.nodes.add(all\_nodes) once. Batch insertion serialises the entire list in a single NDJSON stream, reducing round-trips by an order of magnitude. Similarly, batch-add edges after all nodes are inserted.

**Problem 3:** AttributeError: 'NoneType' object has no attribute 'id' when reading a node ID immediately after creating it

**Cause:** The node ID is assigned by NTE *after* topo.nodes.add() is called. Capturing node.id before the node has been added returns None.

**Solution:** Call topo.nodes.add([node]) before reading node.id. For batch adds, node IDs are set in place on all objects in the list by the time add() returns.

**Problem 4:** Layer derivation does not reflect nodes added after `layers.build()`

**Cause:** Layer derivation is not reactive. Nodes added to the parent layer after `build()` is called are not automatically cloned to derived layers.

**Solution:** Re-run `topo.layers.build(LAYERS)` after any structural change to the parent layer. Freeze the parent layer before building to prevent accidental post-build mutations.

**Problem 5:** Generated configuration is empty or missing protocol sections

**Cause:** The compiler for the selected platform does not have a Jinja2 template for the required protocol (e.g., no IS-IS template for `cisco_ios`).

**Solution:** Run the `operational.platform_support` design rule to list missing template coverage. Either add a custom template under `templates/<platform>/` or switch to a platform compiler that supports the required protocol. See Section 8 for the per-platform protocol coverage table.

**Problem 6:** `LayerValidationError: unknown parent layer 'physical' when calling layers.build()`

**Cause:** The `LAYERS` dictionary references "physical" as a parent, but no nodes have been added to a layer named "physical" yet.

**Solution:** Add nodes with `layer="physical"` before calling `layers.build()`. Alternatively, if using the whiteboard workflow, add a `"physical": {"parent": "whiteboard"}` entry to the `LAYERS` dictionary to derive the physical layer first.

## 11 Integration with the Ecosystem

AutoNetKit sits between the network engineer and a broader automation ecosystem. The following tools interoperate directly.

### 11.1 NTE – Graph Storage and Queries

NTE (`ank_nte`) is the Rust backend that AutoNetKit delegates to for all graph storage and columnar queries. It is installed automatically as a dependency; you do not need to interact with it directly.

For advanced use cases (custom Polars queries, direct graph traversal, petgraph algorithms), import the NTE handle from the topology store:

```
nte_handle = topo._store.ntc # Rust NTE object (PyO3)
df = nte_handle.node_dataframe(layer="physical") # pl.DataFrame
```

**See also:** *NTE Technical Reference* (NTE-TR-001) for engine internals: graph storage (`petgraph::StableDiGraph`), Polars columnar tables, and PyO3 binding details.

### 11.2 NetCfg – Rust-Native Configuration Compiler

NetCfg is an alternative, high-performance configuration compiler written in Rust. Export from AutoNetKit and pass to NetCfg:

```
python -c "
import autonetkit; t = autonetkit.io.load('topology.ndjson')
autonetkit.io.export_json(t, 'topology.json')
"
netcfg compile topology.json --platform junos --output out/
```

**See also:** *NetCfg User Manual* (NETCFG-UM-001) for compilation targets, platform coverage, and template authoring.

### 11.3 TopoGen — Seed-Deterministic Topology Generation

TopoGen generates large, parameterised topologies that AutoNetKit can ingest as a starting point for design validation and config generation:

```
from topo import generate, TopologySpec
from autonetkit.io import from_topogen

spec = TopologySpec(nodes=128, seed=42, topology="partial_mesh")
gen = generate(spec)
topo = from_topogen(gen) # Returns a populated Topology object
```

**See also:** *TopoGen User Manual* (TOPOGEN-UM-001) for seed parameters, topology templates, and large-scale generation strategies.

### 11.4 netsim / netlab — Network Simulation

AutoNetKit exports Containerlab and netsim topology files directly. Use the `NetsimExporter` to generate simulation input and then run netlab to instantiate the lab:

```
python -c "
from autonetkit.batteries_included import isp_topology
from autonetkit.export import NetsimExporter
t = isp_topology(cores=2, edges=4, asn_base=65000)
NetsimExporter().export(t, 'sim/')
"
cd sim && netlab up
```

**See also:** *netsim User Manual* for node image configuration and supported platform capabilities in simulation.

### 11.5 NetVis — Topology Visualisation

AutoNetKit exports GraphML for immediate ingestion by NetVis:

```
topo.io.to_graphml("topology.graphml")
# Open in NetVis or: netvis render topology.graphml
```

**See also:** *NetVis User Manual* (NETVIS-UM-001) for layout algorithms, layer overlays, and export formats.

### 11.6 Workbench — Orchestration

Workbench is the orchestration layer that drives the full design-to-deployment pipeline: it calls AutoNetKit for modelling, NetCfg for compilation, and netsim or Ansible for deployment. Individual AutoNetKit scripts can be wrapped as Workbench tasks:

```
# workbench.yml
tasks:
  - name: build_topology
    type: autonetkit
    script: build_isp.py
    output: topology.ndjson

  - name: compile_configs
    type: netcfg
    input: topology.ndjson
    platform: iosxr
    output: configs/
```

**See also:** *Workbench User Manual* (WB-UM-001) for task graph configuration, secrets management, and CI/CD integration.

## 12 Further Reading

- **ANK-TR-001** — *AutoNetKit: A Type-Safe Framework for Network Topology Modelling*. Architecture deep-dive: generic Pydantic data model, manager-first façade, layer mechanics, query expression engine, rule composition operators, and the configuration generation pipeline. Required reading before extending AutoNetKit with custom managers or compilers.
- **NTE-TR-001** — *NTE: Network Topology Engine Technical Reference*. Rust engine internals: petgraph::StableDiGraph storage, Polars columnar property tables, NDJSON serialisation protocol, and PyO3 binding design.
- **NETCFG-UM-001** — *NetCfg User Manual*. Rust-native configuration compiler: platform registry, Jinja2 template authoring, and batch compilation strategies.
- **TOPOGEN-UM-001** — *TopoGen User Manual*. Seed- deterministic topology generation: parameter reference and integration with AutoNetKit.
- **NETVIS-UM-001** — *NetVis User Manual*. Topology visualisation: GraphML import, interactive layer overlays, and SVG/PNG export.
- **WB-UM-001** — *Workbench User Manual*. End-to-end orchestration: task graphs, AutoNetKit integration, and deployment pipelines.